Datalogger

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Introduktion

Problem

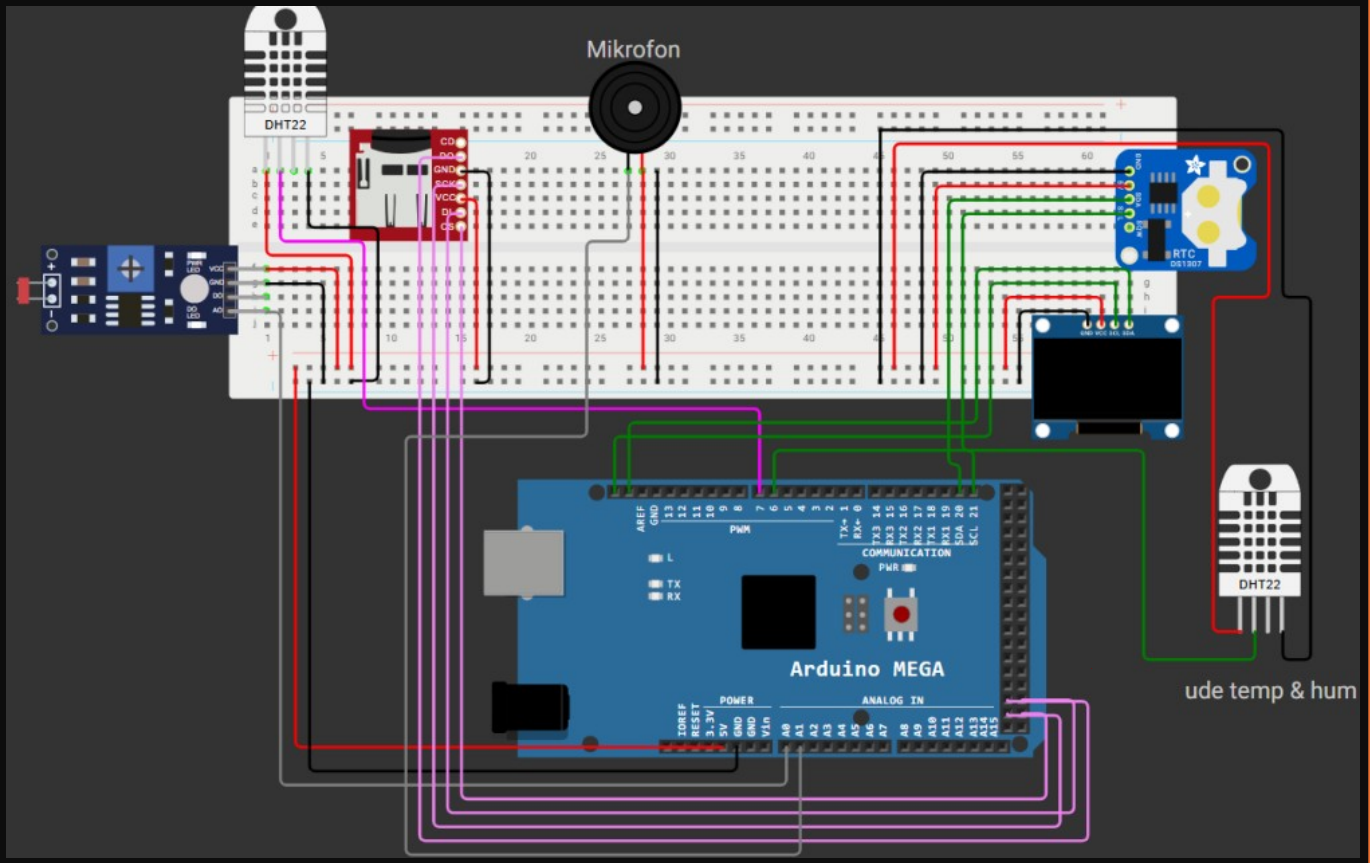
Løsning

Software

Struktur

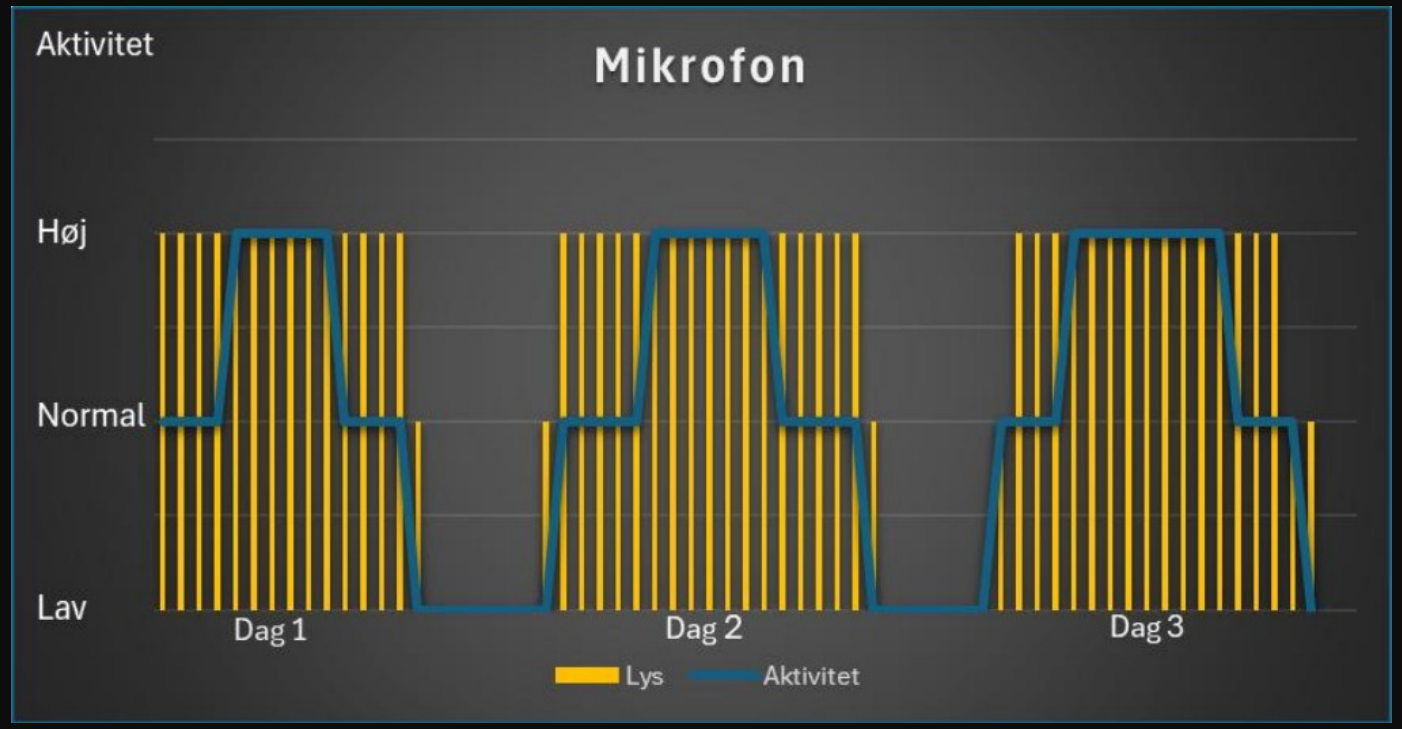
Komponenter

- Arduino Mega – central styreenhed
- DHT22 (inde/ude) – temperatur og fugt (digital)
- LDR – lys (analog via ADC)
- Mikrofon – aktivitet (analog via ADC)



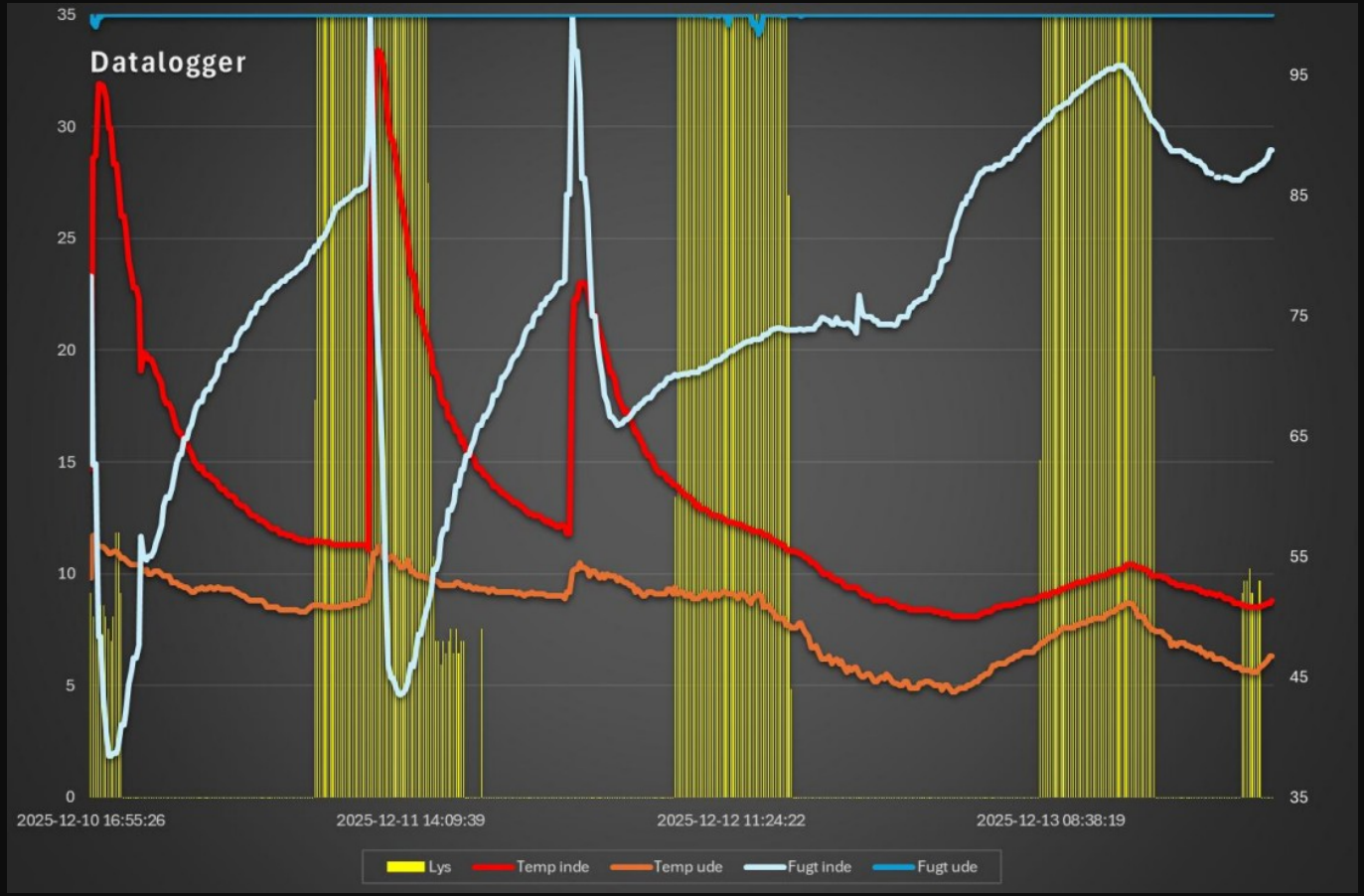
Mikrofon & aktivitet

- Amplitude = max − min
- Klassificeres: lav / normal / høj
- Sammenholdes med lys (dag/nat)



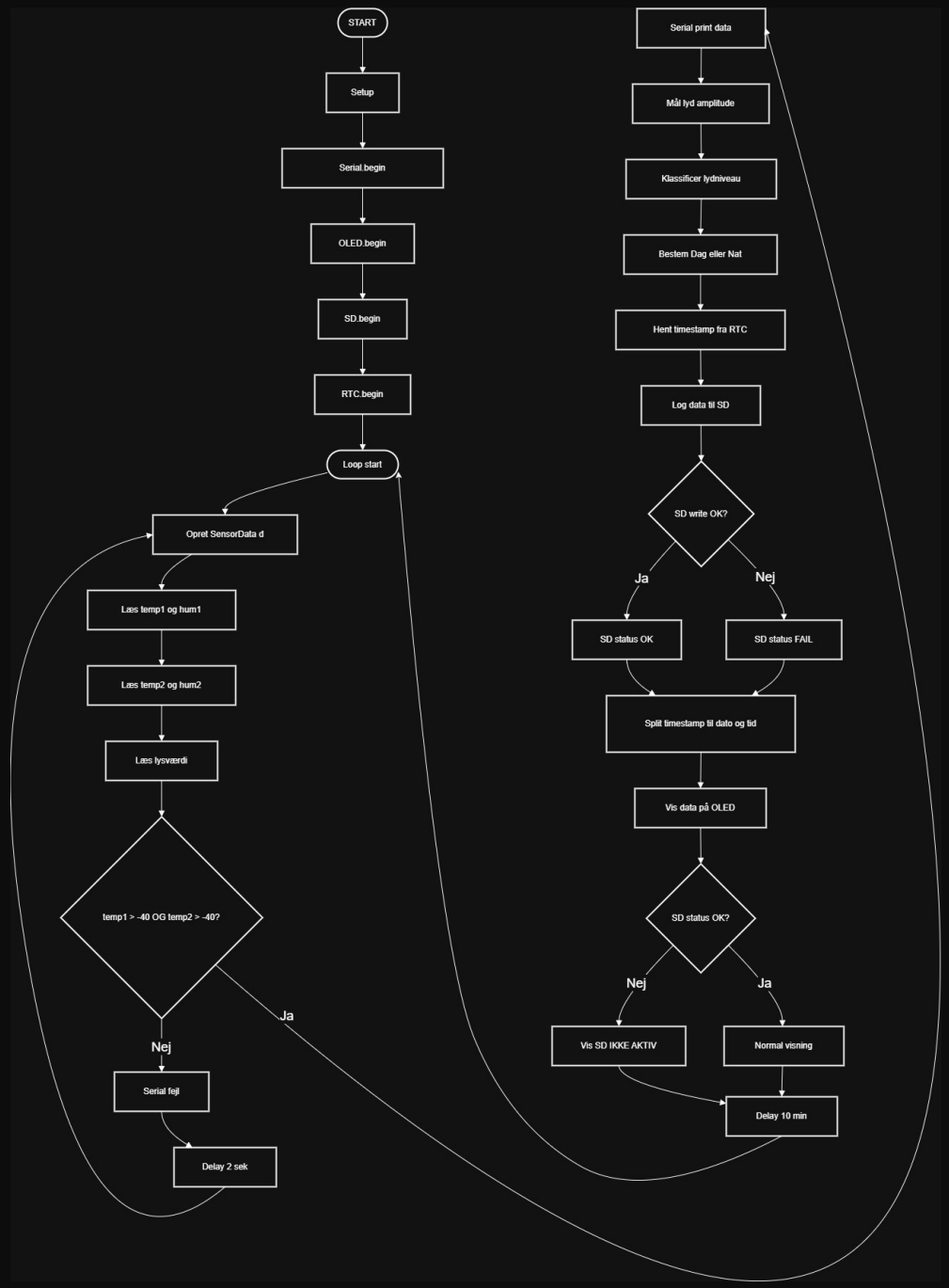
Databehandling

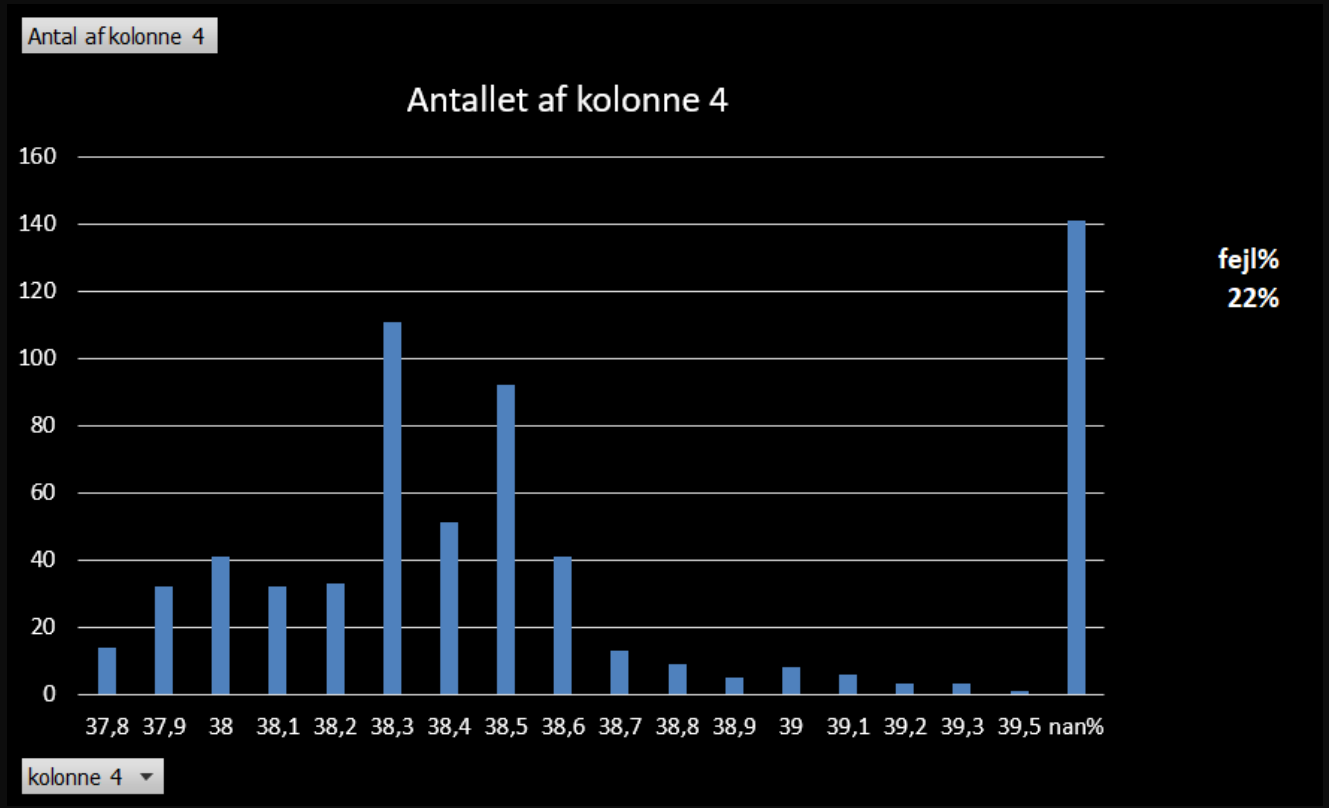
- CSV med tidsstemplede målinger
- Temperatur, fugt og lys over tid
- Bruges til trend-analyse



Kode

- Setup
- Hovedloop
- Sensor aflæsning
- Fejlsikring
- Klassificering
- OLED print
- SD-kort logning
- Delay



Optimering, læring og forbedringer

- Ustabile DHT22 målinger
- Datavalidering med retry
- RTC-fejl
- OLED brugt til lokal status og fejldiagnose
- Forbedringer
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